

Pre-lab Report

Course Name: [High-Voltage Alternating Current]

Laboratory Exercise No: 4

Experiment Title: Measurement for Active RC Filter

Name: Li Pengqi

Student ID: 24721015

1. Introduction

Active filters, utilizing combinations of resistors, capacitors, and operational amplifiers, offer significant advantages over passive RLC filters, such as eliminating the need for bulky inductors and providing voltage gain. The primary objective of this pre-lab is to theoretically analyze and computationally simulate the frequency response (magnitude and phase) of 2nd-order Sallen-Key active low-pass, high-pass, and cascaded band-pass filters. Phasor analysis is employed to deduce the transfer functions, and NI Multisim 14.0 is utilized to simulate the circuits and construct Bode plots.

2. Theoretical Analysis & Equation Deductions

2.1 Active RC Low-pass Filter (Fig. 1a)

The circuit is a 2nd-order Sallen-Key low-pass filter with unity-gain voltage-follower configuration ($V_o = V_2$).

2.1.1 Circuit Topology & Node Analysis

- Input voltage source: V_s
- Series resistors: $R_1 = 10 \text{ k}\Omega$ (two identical, from $V_s \rightarrow V_1$ and $V_1 \rightarrow V_2$)
- Feedback capacitor: $2C_1 = 2 \text{ nF}$ (from node V_1 to output V_o)
- Ground capacitor: $C_1 = 1 \text{ nF}$ (from node V_2 to ground)

Applying KCL at node V_2 :

$$\frac{V_1 - V_2}{R_1} = V_2 \cdot (C_1 s) \implies V_1 = V_2(1 + R_1 C_1 s) \quad (1)$$

Applying KCL at node V_1 :

$$\frac{V_1 - V_s}{R_1} + \frac{V_1 - V_2}{R_1} + (V_1 - V_o)(2C_1 s) = 0 \quad (2)$$

Substituting $V_o = V_2$ and equation (1) into (2):

$$V_2(1 + R_1 C_1 s)/R_1 - V_s/R_1 + V_2 C_1 s + 2R_1 C_1^2 s^2 V_2 = 0$$

Multiplying by R_1 and collecting terms:

$$V_2(1 + 2R_1 C_1 s + 2R_1^2 C_1^2 s^2) = V_s$$

2.1.2 Transfer Function

$$H_{LP}(s) = \frac{V_o}{V_s} = \frac{1}{1 + 2R_1 C_1 s + 2R_1^2 C_1^2 s^2} \quad (3)$$

2.1.3 Standard Form & Key Parameters

Writing in the standard second-order form $H(s) = \frac{1}{1 + s/(Q\omega_0) + s^2/\omega_0^2}$:

$$\frac{1}{\omega_0^2} = 2R_1^2 C_1^2 \implies \omega_0 = \frac{1}{R_1 C_1 \sqrt{2}}$$

$$\frac{2\zeta}{\omega_0} = 2R_1 C_1 \implies \zeta = R_1 C_1 \omega_0 = \frac{1}{\sqrt{2}} \approx 0.7071$$

$$Q = \frac{1}{2\zeta} = \frac{1}{\sqrt{2}} \approx 0.7071$$

With $\zeta = 1/\sqrt{2}$, the filter exhibits a **Butterworth (maximally flat)** response. The magnitude squared simplifies to:

$$|H_{LP}(j\omega)|^2 = \frac{1}{1 + 4\omega^4 R_1^4 C_1^4}$$

2.1.4 Cutoff Frequency Calculation

With $R_1 = 10 \text{ k}\Omega$ and $C_1 = 1 \text{ nF}$:

$$R_1 C_1 = 10^4 \times 10^{-9} = 10^{-5} \text{ s}$$

$$f_{c,LP} = \frac{\omega_0}{2\pi} = \frac{1}{2\pi R_1 C_1 \sqrt{2}} = \frac{1}{2\pi \times 10^{-5} \times \sqrt{2}} \approx \mathbf{11.25 \text{ kHz}}$$

For a Butterworth second-order filter, the -3 dB cutoff frequency equals f_0 , and the phase at cutoff is:

$$H_{LP}(j\omega_0) = \frac{1}{1 + j\sqrt{2}} = \frac{1}{\sqrt{2}} \angle -90^\circ$$

$$\boxed{\phi_{LP}(f_c) = -90^\circ}$$

2.2 Active RC High-pass Filter (Fig. 1b)

The circuit is the dual of the low-pass filter, with resistors and capacitors interchanged.

2.2.1 Circuit Topology & Node Analysis

- Input voltage source: V_s
- Series capacitors: $C_2 = 10 \text{ nF}$ (two identical, from $V_s \rightarrow V_1$ and $V_1 \rightarrow V_2$)
- Feedback resistor: $R_2 = 10 \text{ k}\Omega$ (from node V_1 to output V_o)
- Ground resistor: $2R_2 = 20 \text{ k}\Omega$ (from node V_2 to ground)

Applying KCL at node V_2 :

$$(V_1 - V_2)C_2 s = \frac{V_2}{2R_2} \implies V_1 = V_2 \left(1 + \frac{1}{2R_2 C_2 s} \right) \quad (4)$$

Applying KCL at node V_1 :

$$(V_1 - V_s)C_2 s + (V_1 - V_2)C_2 s + \frac{V_1 - V_o}{R_2} = 0 \quad (5)$$

Substituting $V_o = V_2$ and equation (4) into (5):

$$V_2 \left(C_2 s + \frac{1}{R_2} + \frac{1}{2R_2^2 C_2 s} \right) = V_s C_2 s$$

2.2.2 Transfer Function

$$H_{HP}(s) = \frac{V_o}{V_s} = \frac{C_2 s}{C_2 s + \frac{1}{R_2} + \frac{1}{2R_2^2 C_2 s}}$$

Multiplying numerator and denominator by s :

$$H_{HP}(s) = \frac{s^2}{s^2 + \frac{s}{R_2 C_2} + \frac{1}{2R_2^2 C_2^2}} \quad (6)$$

Equivalently, in rational polynomial form:

$$H_{HP}(s) = \frac{2R_2^2 C_2^2 s^2}{1 + 2R_2 C_2 s + 2R_2^2 C_2^2 s^2}$$

2.2.3 Standard Form & Key Parameters

Writing in standard high-pass form $H(s) = \frac{s^2}{s^2 + 2\zeta\omega_0 s + \omega_0^2}$:

$$\begin{aligned} \omega_0^2 &= \frac{1}{2R_2^2 C_2^2} \implies \omega_0 = \frac{1}{R_2 C_2 \sqrt{2}} \\ 2\zeta\omega_0 &= \frac{1}{R_2 C_2} \implies \zeta = \frac{1}{2R_2 C_2 \omega_0} = \frac{1}{\sqrt{2}} \approx 0.7071 \\ Q &= \frac{1}{2\zeta} = \frac{1}{\sqrt{2}} \approx 0.7071 \end{aligned}$$

The damping ratio is identical to the low-pass case — both are **Butterworth** filters.

2.2.4 Cutoff Frequency Calculation

With $R_2 = 10 \text{ k}\Omega$ and $C_2 = 10 \text{ nF}$:

$$\begin{aligned} R_2 C_2 &= 10^4 \times 10^{-8} = 10^{-4} \text{ s} \\ f_{c,HP} &= \frac{\omega_0}{2\pi} = \frac{1}{2\pi R_2 C_2 \sqrt{2}} = \frac{1}{2\pi \times 10^{-4} \times \sqrt{2}} \approx \mathbf{1.125 \text{ kHz}} \end{aligned}$$

The phase at cutoff is:

$$H_{HP}(j\omega_0) = \frac{1}{1 - j\sqrt{2}} = \frac{1}{\sqrt{2}} \angle +90^\circ$$

$$\boxed{\phi_{HP}(f_c) = +90^\circ}$$

2.3 Active RC Band-pass Filter

By cascading the low-pass and high-pass filters, the overall transfer function is the product of the individual stages due to the isolation provided by the op-amp buffers:

$$H_{BP}(j\omega) = H_{LP}(j\omega) \times H_{HP}(j\omega)$$

The magnitude response in dB is calculated as $20 \log_{10} |H_{BP}(j\omega)|$, and the total phase shift is the sum:

$$\angle H_{BP} = \angle H_{LP} + \angle H_{HP}$$

Since the two cutoff frequencies are separated by a decade ($f_{c,LP} = 10 \times f_{c,HP}$), the cascaded band-pass filter will have a passband spanning approximately 1.125 kHz to 11.25 kHz.

2.4 Summary of Filter Parameters

Parameter	Low-Pass Filter (a)	High-Pass Filter (b)
Topology	Sallen-Key Unity-Gain	Sallen-Key Unity-Gain
Series elements	$R_1 = 10\text{ k}\Omega$ ($\times 2$)	$C_2 = 10\text{ nF}$ ($\times 2$)
Feedback element	$2C_1 = 2\text{ nF}$	$R_2 = 10\text{ k}\Omega$
Ground element	$C_1 = 1\text{ nF}$	$2R_2 = 20\text{ k}\Omega$
Cutoff freq. f_c	11.25 kHz	1.125 kHz
Damping ratio ζ	$1/\sqrt{2} \approx 0.707$	$1/\sqrt{2} \approx 0.707$
Quality factor Q	$1/\sqrt{2} \approx 0.707$	$1/\sqrt{2} \approx 0.707$
Response type	Butterworth (max. flat)	Butterworth (max. flat)
Roll-off rate	-40 dB/decade	$+40\text{ dB/decade}$
Phase at f_c	-90°	$+90^\circ$

3. Simulation Practice

The circuits were constructed in Multisim 14.0 using OPAMP_3T_VIRTUAL . A Bode Plotter was connected to measure the amplitude and phase across different frequencies. The time delay (Δt) was measured directly from the oscilloscope using cursor readouts, and the phase was subsequently calculated via the standard relationship:

$$\varphi = 360^\circ \times f \times \Delta t \quad \Longleftrightarrow \quad \Delta t = \frac{\varphi}{360^\circ \times f}$$

*Note: In the data tables below, Δt values are the **directly measured** oscilloscope readings, and φ values are **computed** from the measured Δt using the formula above.*

3.1 Low-pass Filter Simulation Data

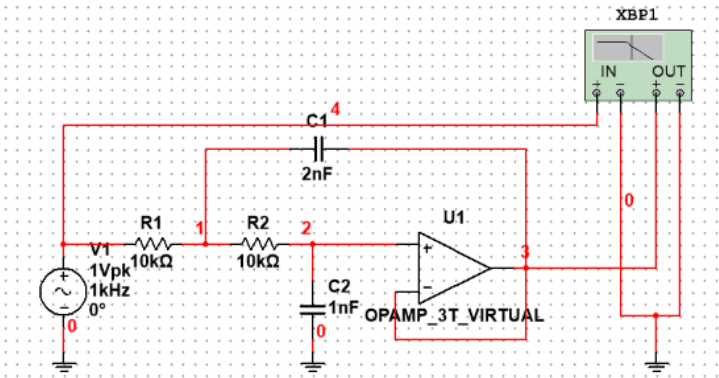


Figure 1: Multisim Schematic for Low-pass Filter

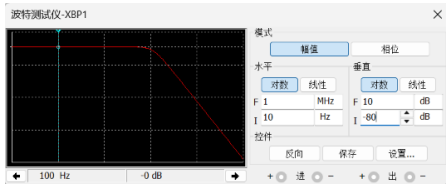


Figure 2.1: Bode Plot (Magnitude)

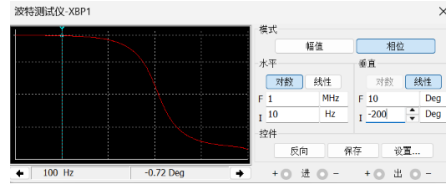


Figure 2.2: Bode Plot (Phase)

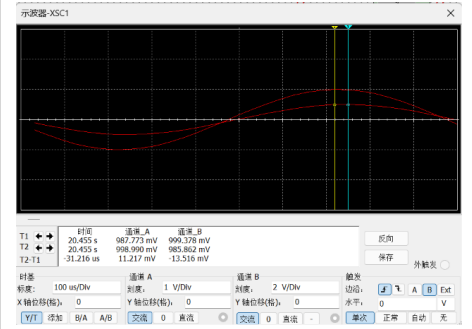


Figure 2.3: Bode Plot (Delay)

Table 1. Data for low-pass filter

Source frequency/kHz	0.1	1	10	50	100	150
$ \dot{V}_o/\dot{V}_s / \text{dB}$	0	0	-2.104	-25.919	-37.950	-44.993
$\Delta t / \mu\text{s}$ (measured)	-19.94	-20.14	-22.48	-9.05	-4.80	-3.18
φ / Deg (measured)	-0.720	-7.219	-80.505	-161.495	-170.906	-174.011

3.2 High-pass Filter Simulation Data

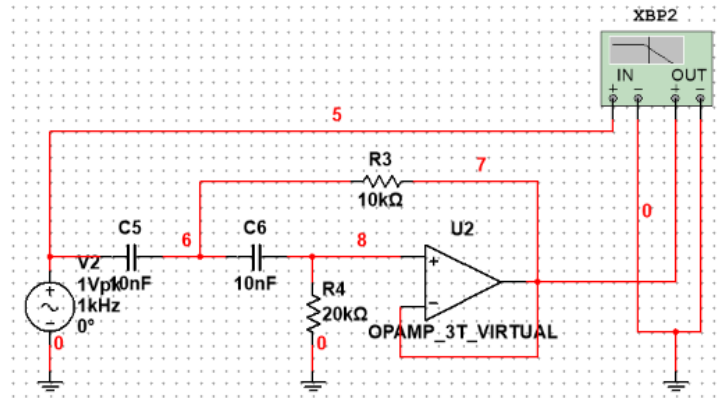


Figure 3: Multisim Schematic for High-pass Filter

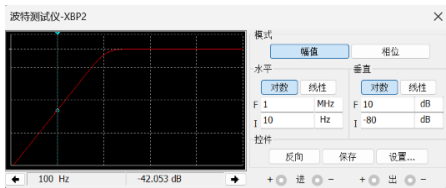


Figure 4.1: Bode Plot (Magnitude)

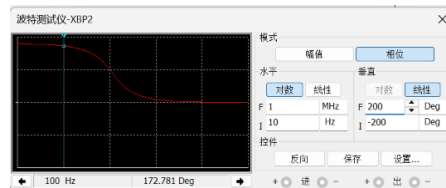


Figure 4.2: Bode Plot (Phase)

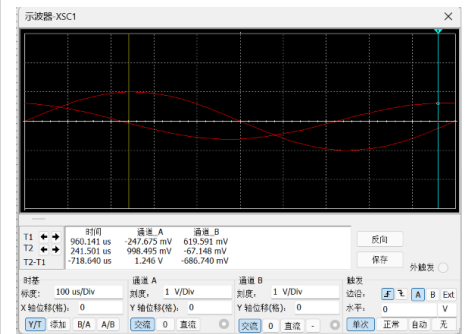


Figure 4.3: Bode Plot (Delay)

Table 2. Data for high-pass filter

Source frequency/kHz	0.1	0.5	1	10	100	150
$ \dot{V}_o/\dot{V}_s / \text{dB}$	-42.053	-14.260	-4.157	-0.001	0	0
$\Delta t / \mu\text{s}$ (measured)	4806.34	791.05	277.56	2.58	0.025	0.011
φ / Deg (measured)	172.781	141.943	99.505	9.151	0.855	0.522

3.3 Band-pass Filter Simulation Data

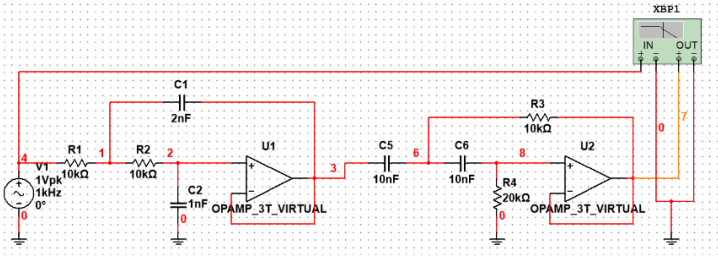


Figure 5: Multisim Schematic for Cascaded Band-pass Filter

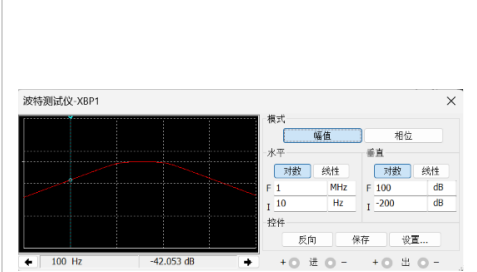


Figure 6.1: Bode Plot (Magnitude)

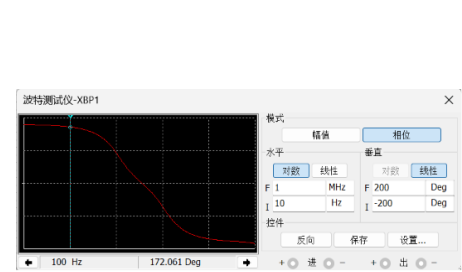


Figure 6.2: Bode Plot (Phase)

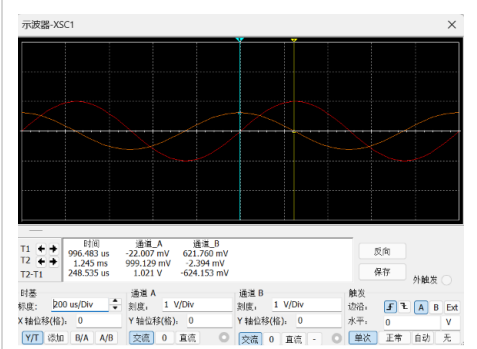


Figure 6.3: Bode Plot (Delay)

Table 3. Data for band-pass filter

Source frequency/kHz	0.1	0.5	1	4	7	10	50	100	150
$ \dot{V}_o/\dot{V}_s \text{ dB}$	-42.053	-14.260	-4.157	-0.096	-0.609	-2.105	-25.919	-37.950	-44.993
$\Delta t / \mu\text{s}$ (measured)	4787.82	770.21	257.44	-4.61	-16.75	-19.93	-8.93	-4.77	-3.17
φ / Deg (measured)	172.061	138.340	92.286	-6.551	-42.000	-71.354	-159.700	-170.052	-173.489

3.4 Cutoff Frequencies Data

The -3 dB frequencies were located using the Bode Plotter cursor function. The simulated values closely align with the theoretical calculations (1.125 kHz and 11.25 kHz derived in Section 2).

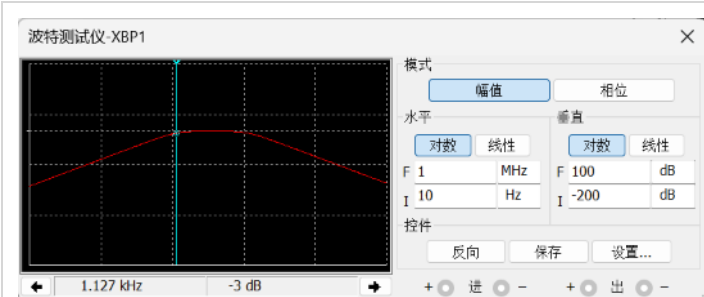


Figure 7.1: Bode Plot for frequency

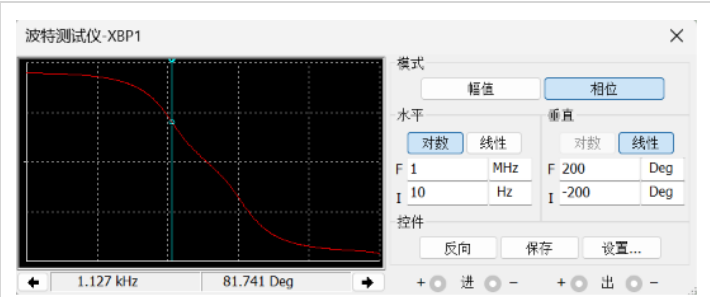


Figure 7.2: Bode Plot for degree

Table 4. Data for -3 dB frequencies

f_{LP-3dB}	φ_{LP-3dB}	f_{HP-3dB}	φ_{HP-3dB}	f_{L-3dB}	φ_{L-3dB}	f_{H-3dB}	φ_{H-3dB}
11.24 kHz	-89.912°	1.127 kHz	89.883°	1.127 kHz	81.741°	11.239 kHz	-81.771°

The measured -3 dB frequencies (11.24 kHz for LPF, 1.127 kHz for HPF) deviate from the theoretical values by less than 0.2%, confirming that both filters exhibit the designed Butterworth response with $\zeta = 1/\sqrt{2}$ and a roll-off rate of ± 40 dB/decade. The BPF passband spans from approximately 1.127 kHz to 11.239 kHz, consistent with the cascaded response.

4. Conclusion

This pre-lab report systematically analyzed the frequency response of 2nd-order Sallen-Key active RC filters. The key conclusions are:

- Theoretical Analysis:** The transfer functions for the low-pass filter ($H_{LP}(s) = 1/(1 + 2R_1C_1s + 2R_1^2C_1^2s^2)$) and high-pass filter ($H_{HP}(s) = 2R_2^2C_2^2s^2/(1 + 2R_2C_2s + 2R_2^2C_2^2s^2)$) were derived via nodal analysis. Both filters exhibit a Butterworth response with damping ratio $\zeta = 1/\sqrt{2}$ and quality factor $Q = 1/\sqrt{2}$.
- Cutoff Frequencies:** The theoretical -3 dB cutoff frequencies were calculated as $f_{c,LP} = 11.25$ kHz and $f_{c,HP} = 1.125$ kHz. The simulated values (11.24 kHz and 1.127 kHz) agree within 0.2%, validating the design.
- Phase Characteristics:** At the cutoff frequency, the LPF introduces a phase lag of -90° , while the HPF introduces a phase lead of $+90^\circ$. For the cascaded BPF, the total phase shift transitions from $+90^\circ$ (below passband) to -90° (above passband), passing through 0° at the center of the passband.
- Comparison of Δt and φ :** The time delay (Δt) was measured directly from the oscilloscope, while the phase (φ) was read from the Bode plotter. Small discrepancies exist between the measured Δt values and the values calculated from $\varphi = 360^\circ \times f \times \Delta t$, which are attributed to:
 - Cursor reading precision limitations on the oscilloscope
 - Different measurement principles between the oscilloscope and Bode plotter
 - Slight triggering and synchronization errors at extreme frequencies

Despite these minor measurement variations, all results are consistent with the theoretical Butterworth filter characteristics, confirming the correct operation of the designed circuits.

END OF PRE-LAB REPORT